Chenmu Zhang

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RESEARCH INTERESTS

- **Agentic AI for materials science:** autonomous simulation and discovery; agent-driven model development; evaluation of scientific reasoning and research workflows.
- **Multi-agent scientific research:** parallel exploration, knowledge sharing and coordination; effects of collaboration on research progress.
- **Machine learning for electronic structure and dynamics:** learned Hamiltonians, interatomic potentials and surrogates for first-principles simulations.

RESEARCH EXPERIENCE

► **Postdoctoral Research Associate** May 2023 – Present

Rice University, Houston, TX Advisor: Prof. Boris I. Yakobson

- Designed and built **MatClaw**, an open-source LLM agent that writes and executes code to carry out data analysis and long-running materials simulations on HPC clusters.
- Applied MatClaw to study the energetic and electronic effects of off-stoichiometry in Cu–In–Se, linking generative structure prediction, interatomic-potential screening and DFT validation (co-first-author manuscript in preparation).
- Applied autonomous research to band-gap model development: an agent redesigned, trained and evaluated models over **184 experiments with no human in the loop**. The resulting model, trained without external pretraining, outperformed 17 published expert-designed models in the same category, based on their reported MatBench scores (LM4Sci workshop, COLM 2026).
- Led the design and agent evaluations for **CompMat-Bench**: 94 tasks from published materials studies, graded against reproduced results. Compared agents across three LLMs on individual tasks and multi-step workflows to assess scientific research capability.
- Investigated stochastic defect dynamics as an entropy source for true random bits in 2D semiconductors, and defect-controlled thermal transport in ZrC at high temperatures (two manuscripts submitted).

► **Graduate Research Assistant** Aug. 2018 – Apr. 2023

University of Texas at Austin, Austin, TX Advisor: Prof. Yuanyue Liu

- Contributed to explaining the low carrier mobility of 2D semiconductors (PRL 2020); identified electronic structures supporting high room-temperature mobility (PRL 2023).
- Extended first-principles electron-phonon transport to quadrupole, substrate, surface and remote-phonon scattering; identified mobility enhancement by remote phonons (PRB 2022; ACS Nano 2024; Nano Letters 2026).

EDUCATION

► **University of Texas at Austin, Austin, TX** Aug. 2018 – Apr. 2023

Ph.D. in Materials Engineering
Advisor: Prof. Yuanyue Liu
Thesis: *"Phonon-limited Carrier Transport in 2D Materials from First Principles"*

► **University of Science and Technology of China (USTC), Hefei, China** Sep. 2014 – Jul. 2018

B.S. in Physics
Advisor: Prof. Wenguang Zhu
Thesis: *"First Principles Calculations on 2D Ferroelectric Materials under Uniaxial Strain"*

PUBLICATIONS

13 peer-reviewed publications (8 first-author), including 12 journal articles and one accepted workshop paper; 3 manuscripts under review. 613 citations, h-index 10 (Google Scholar, Aug. 2026). [Full profile]

Peer-reviewed Publications

1. **Zhang, C.**; Yakobson, B. I. Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. arXiv preprint arXiv:2606.29717, accepted at the Workshop on Language Models for Scientific Discovery (LM4Sci), COLM **2026** [Paper].
2. **Zhang, C.**; Liu, Y. Enhancing Carrier Mobility by Remote Phonons. *Nano Letters* **2026**, 26(30), 9716–9721 [Paper].
3. **Zhang, C.**; Xiao, Z.; Paddock, R.; Cullinan, M.; Tehrani, M.; Liu, Y. Effects of Graphene Doping on the Electrical Conductivity of Copper. *Advanced Functional Materials* **2024**, 34(45), 2407569 [Paper].
4. **Zhang, C.**; Liu, Y. Electron-Surface Scattering from First-Principles. *ACS Nano* **2024**, 18(40), 27433–27439 [Paper].
5. Xiao, Z.; Guo, R.; **Zhang, C.**; Liu, Y. Point Defect Limited Carrier Mobility in 2D Transition Metal Dichalcogenides. *ACS Nano* **2024**, 18(11), 8511–8516 [Paper].
6. **Zhang, C.**; Wang, R.; Mishra, H.; Liu, Y. Two-Dimensional Semiconductors with High Intrinsic Carrier Mobility at Room Temperature. *Physical Review Letters* **2023**, 130(8), 087001 [Paper].
7. **Zhang, C.**; Liu, Y. Phonon-limited transport of two-dimensional semiconductors: Quadrupole scattering and free carrier screening. *Physical Review B* **2022**, 106(11), 115423 [Paper].
8. Cheng, L.; **Zhang, C.**; Liu, Y. Intrinsic charge carrier mobility of 2D semiconductors. *Computational Materials Science* **2021**, 194, 110468 [Paper].
9. **Zhang, C.**; Cheng, L.; Liu, Y. Role of flexural phonons in carrier mobility of two-dimensional semiconductors: free standing vs on substrate. *Journal of Physics: Condensed Matter* **2021**, 33(23), 234003 [Paper].
10. Cheng, L.; **Zhang, C.**; Liu, Y. Why Two-Dimensional Semiconductors Generally Have Low Electron Mobility. *Physical Review Letters* **2020**, 125(17), 177701 [Paper].
11. **Zhang, C.**; Cheng, L.; Liu, Y. Understanding high-field electron transport properties and strain effects of monolayer transition metal dichalcogenides. *Physical Review B* **2020**, 102(11), 115405 [Paper].
12. Cheng, L.; **Zhang, C.**; Liu, Y. How to resolve a phonon-associated property into contributions of basic phonon modes. *Journal of Physics: Materials* **2019**, 2(4), 045005 [Paper].
13. Cheng, L.; **Zhang, C.**; Liu, Y. The Optimal Electronic Structure for High-Mobility 2D Semiconductors: Exceptionally High Hole Mobility in 2D Antimony. *Journal of the American Chemical Society* **2019**, 141(41), 16296–16302 [Paper].

Preprints and Submitted Manuscripts

14. **Zhang, C.**; Felix, L. C.; Zhang, J.-J.; Li, X.; Jiang, X.; Jiang, T.; Mishra, S.; Qin, X.; Yakobson, B. I. CompMat-Bench: Benchmarking AI Agents for Computational Materials Science. arXiv preprint arXiv:2610.00636, (Submitted) **2026** [Paper].
15. **Zhang*, C.**; Felix*, L. C.; Yakobson, B. I. Defect-controlled thermal conductivity of ZrC at high temperatures from first principles. (Submitted; * equal contribution) **2026**.
16. **Zhang, C.**; Sun, F.; Yakobson, B. I. Design of efficient true random bits generation in transition metal dichalcogenides from first principles. (Submitted) **2026**.
17. **Zhang, C.**; Yakobson, B. I. MatClaw: An Autonomous Code-First LLM Agent for End-to-End Materials Exploration. arXiv preprint arXiv:2604.02688, **2026** [Paper].
18. **Zhang, C.**; Liu, Y. Two-dimensional Semiconductor Computational Carrier Mobility Genome. arXiv preprint arXiv:2404.08117, **2024** [Paper].

Manuscripts in Preparation

19. Yuan*, Z.; **Zhang*, C.** Off-Stoichiometric Energetic and Electronic Asymmetry in Ternary Cu–In–Se: Task-Oriented Materials Discovery by an Autonomous LLM Agent. (In preparation; * equal contribution) **2026**.

RESEARCH SOFTWARE AND DATASETS

- **MatClaw** [Code] [Paper] — autonomous agent for computational materials research, with persistent Python state, agent-written tools and remote HPC execution. Open source; sole developer.
- **CompMat-Bench** [Code and Data] [Paper] — 94 tasks and 10 workflows from 15 published studies, with fixed-rule grading against reproduced results. Released code, task data and 976 agent trajectories; led benchmark design and agent evaluations.
- **Autonomous LLM Research for Materials Property Prediction** [Code] [Paper] — agent-driven model redesign, training and evaluation for band-gap prediction on MatBench. Built and applied the autonomous research loop.
- **qeh-frohlich** [Code] [Paper] — phonon-limited mobility in van der Waals heterostructures with Fröhlich coupling; the code behind the remote-phonon study (Nano Lett. 2026). Sole developer.
- **EDI** [Code] — first-principles electron-defect scattering and defect-limited transport. Contributor.

PRESENTATIONS

- How Well Can AI Agents Do Computational Materials Research? A Benchmark Built from Peer-Reviewed Publications. MRS Fall Meeting, Symposium MT01, Boston, Dec. 2026 (talk, scheduled).
- Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. LM4Sci Workshop at COLM 2026, San Francisco, Oct. 2026 (poster).
- Autonomous Materials Simulation with an LLM Agent: Modeling Exothermic Fronts in 2D Ferroelectrics. APS Global Physics Summit, Denver, 2026 (talk).
- First-Principles Electron-Surface Scattering: Case Study on Copper Films. APS Global Physics Summit, Anaheim, 2025 (talk).
- Harnessing Stochasticity in 2D Materials: Phonon-Assisted Defect Charge Dynamics in WS_2. APS Global Physics Summit, Anaheim, 2025 (poster).
- Electron-Surface Scattering from First Principles. APS March Meeting, Minneapolis, 2024 (talk).
- New 2D Semiconductors with Ultra-High Mobility at Room Temperature. APS March Meeting, Minneapolis, 2024 (poster).

PROFESSIONAL SERVICE

- Reviewer for *Physical Review B*, *Ceramics International*, and *Journal of Superconductivity and Novel Magnetism*

TEACHING

- Teaching Assistant, Structure of Materials (Fall 2021, Fall 2022), University of Texas at Austin